

Electric Circuits

Charge, energy and resistance — a twelve-lesson route from first circuits to confident AQA-style answers.

Predict. Measure. Explain. Then prove it with an equation.

NAME

CLASS

TEACHER

LAB PARTNER

Twelve lessons. One connected story.

The moving charge is conserved. Energy is transferred by the components. Measurements turn those ideas into equations, graphs and explanations.

HOW EVERY LESSON WORKS

The case → your mission → a key idea → worked example → numbered tasks → AQA-style checkpoint. Write in the spaces. When a task points to an existing virtual-lab or required-practical sheet, use that sheet alongside this booklet.

THE ROUTE

01 Charge and current	07 Reading I-V graphs
02 Potential difference	08 Required practical: I-V
03 Resistance and Ohm's law	09 Thermistors and LDRs
04 Required practical: wire length	10 Mains and safety
05 Series and parallel	11 Power and energy
06 Required practical: combinations	12 National Grid and exam clinic

WHERE TO FIND THE LIVE RESOURCES

Student companion: panphy.app/gcsephy/year10phy/unit01/

Virtual labs: use the linked 16-page 'Y10 Electricity Virtual Labs' PDF and PhET Circuit Construction Kit DC - Lab.
Required practicals: use the linked Resistance and I-V Characteristics worksheets for full methods, diagrams and extra analysis.

The equation toolkit

Choose the relationship from the quantities in the question. Rearrange before inserting numbers. Check units and whether the answer is sensible.

$$Q = It$$

charge = current x time

$$C = A \times s$$

$$E = QV$$

energy = charge x p.d.

$$J = C \times V$$

$$V = IR$$

p.d. = current x resistance

$$V = A \times \text{ohm}$$

$$R_{\text{total}} = R1 + R2$$

total resistance in series

all in ohm

$$P = VI$$

power = p.d. x current

$$W = V \times A$$

$$P = I^2R$$

power = current squared x resistance

$$W = A^2 \times \text{ohm}$$

$$E = Pt$$

energy = power x time

$$J = W \times s$$

$$V_{\text{pIp}} = V_{\text{sIs}}$$

ideal transformer, HT only

primary power = secondary power

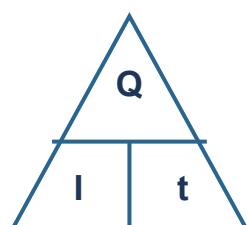
Exam-year note

AQA supplies a physics equations sheet for the 2027 exams. Practise selecting and rearranging equations independently; check the arrangement for the cohort's actual exam year. The transformer relation applies to Higher tier.

Equation triangles

Cover the letter you want. Across the bottom means multiply; top over bottom means divide. For squared current, undo the square root last.

Charge

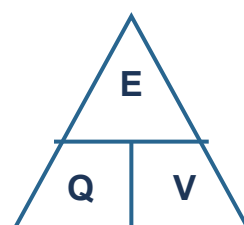


$$Q = I \times t$$

$$I = Q / t$$

$$t = Q / I$$

Electrical work

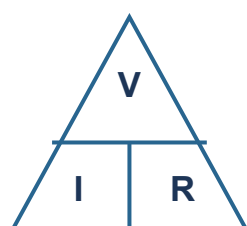


$$E = Q \times V$$

$$Q = E / V$$

$$V = E / Q$$

Resistance

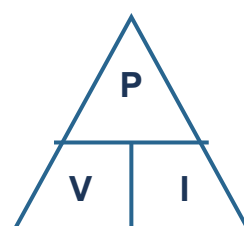


$$V = I \times R$$

$$I = V / R$$

$$R = V / I$$

Electrical power

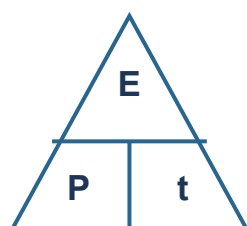


$$P = V \times I$$

$$V = P / I$$

$$I = P / V$$

Energy and time

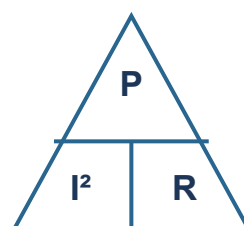


$$E = P \times t$$

$$P = E / t$$

$$t = E / P$$

Heating power



$$P = I^2 \times R$$

$$R = P / I^2$$

$$I = \sqrt{P / R}$$

Not triangle rules: in series, $R_{\text{total}} = R_1 + R_2 + \dots$. For an ideal Higher-tier transformer, $V_p I_p = V_s I_s$; divide the product of the known side by the known quantity on the other side.

Charge on the move

Circuit symbols, current and charge

THE CASE

A lamp goes out when a wire comes loose. The charges were already in the wires: the missing connection stopped the complete path.

YOUR MISSION

Explain what moves in a closed circuit and calculate charge flow.

KEY IDEA

Current is the rate of charge flow. In one series loop, current is the same at every point; components transfer energy, not charge.

WORKED EXAMPLE

A current of 0.40 A flows for 3.0 min. $t = 180 \text{ s}$; $Q = It = 0.40 \times 180 = 72 \text{ C}$.

1 Build the idea

1. Draw a cell, switch, lamp, resistor, ammeter and voltmeter using standard circuit symbols. Add a closed loop containing a cell, switch, lamp and ammeter.

2. A student says current is used up by a lamp. Predict readings immediately before and after it, then explain your prediction.

Measure, reason, answer

2 Use the evidence

3. Virtual Lab 1: use the linked PhET lab and existing worksheet pp. 3-6. Record current at three places in one loop. Change the resistor; record what changes and what stays equal.

4. Calculate charge flow for 0.25 A lasting 4.0 min. Show the time conversion and unit.

3 AQA-style checkpoint

[4 marks] A motor draws 0.60 A for 5.0 minutes. Calculate the charge that passes it and explain whether the current after the motor is smaller than before it.

CHECK BEFORE MOVING ON

Have I shown a formula or rule, used the given data, included units and answered the words in the question?

Energy per coulomb

Potential difference and electrical work

THE CASE

The same charge can give a lamp more energy when the supply p.d. is higher. Voltage describes energy per coulomb, not the speed of the charges.

YOUR MISSION

Explain p.d. as energy transferred per coulomb and calculate $E = QV$.

KEY IDEA

Potential difference is energy transferred per unit charge: $1 \text{ V} = 1 \text{ J/C}$. A voltmeter is connected in parallel across the component.

WORKED EXAMPLE

A lamp transfers 36 J as 6.0 C passes. $V = E/Q = 36/6.0 = 6.0 \text{ V}$.

1 Build the idea

1. Draw a voltmeter across a lamp and an ammeter in series with it. Explain each position.

2. Predict how the p.d. across a single lamp changes when the battery changes from 3 V to 6 V.

Measure, reason, answer

2 Use the evidence

3. Virtual Lab 2: complete the p.d. and energy investigation on existing worksheet pp. 7-11. Record p.d. across the battery and two series components; test whether the drops add.

4. Calculate energy transferred when 25 C passes through a component with 4.0 V across it.

3 AQA-style checkpoint

[4 marks] A 9.0 V battery moves 20 C of charge through a circuit. Calculate the energy transferred by the battery. Explain what a voltmeter reading of 3.0 V across one resistor means.

CHECK BEFORE MOVING ON

Have I shown a formula or rule, used the given data, included units and answered the words in the question?

Resistance and Ohm's law

$V = IR$, resistance and temperature

THE CASE

Two devices on the same supply draw different currents. Their resistances differ: $R = V/I$ compares p.d. with current.

YOUR MISSION

Calculate resistance and explain why a fixed resistor can obey $V = IR$.

KEY IDEA

For a fixed resistor at constant temperature, current is directly proportional to p.d. Resistance is measured in ohms (ohm). Heating can change resistance.

WORKED EXAMPLE

A resistor has 6.0 V across it and 0.30 A through it. $R = V/I = 6.0/0.30 = 20$ ohm.

1 Build the idea

1. Calculate R for (a) 12 V, 0.60 A; (b) 3.0 V, 0.15 A. State the unit.

2. Explain why the phrase 'a resistor obeys Ohm's law' needs the words 'at constant temperature'.

Measure, reason, answer

2 Use the evidence

3. Virtual Lab 3: complete the resistance investigation on existing worksheet pp. 12-16. Change one variable at a time and compare calculated R values.

4. Sketch current (y) against p.d. (x) for a fixed resistor and state what the gradient tells you qualitatively.

3 AQA-style checkpoint

[4 marks] A fixed resistor has 2.0 V across it and carries 0.10 A. Calculate its resistance and predict the current at 6.0 V if its temperature stays constant.

CHECK BEFORE MOVING ON

Have I shown a formula or rule, used the given data, included units and answered the words in the question?

The resistance of a wire

Required practical: length of a wire, Part A

THE CASE

A longer wire gives moving charges more material to pass through. The test must keep its material, diameter and temperature the same.

YOUR MISSION

Plan, collect and graph resistance against wire length while controlling temperature.

KEY IDEA

Change the length between the crocodile clips. Measure V and I , then calculate $R = V/I$. Switch off between readings to limit heating.

WORKED EXAMPLE

At 40 cm, $V = 0.80 \text{ V}$ and $I = 0.40 \text{ A}$. $R = 0.80/0.40 = 2.0 \text{ ohm}$.

1 Build the idea

1. Label independent, dependent and three control variables. State why temperature matters.

2. Draw the circuit: supply, switch, ammeter and test wire in series; voltmeter across only the measured wire length.

Measure, reason, answer

2 Use the evidence

3. Collect 10-100 cm readings in the table. Calculate resistance at each length. Repeat any result that does not fit the pattern.

4. Plot R (y) against length (x), draw a best-fit line and write a conclusion supported by two readings.

3 AQA-style checkpoint

[6 marks] Describe how you would investigate how the resistance of a wire depends on its length. Include the circuit, measurements, controls, safety and processing.

CHECK BEFORE MOVING ON

Have I shown a formula or rule, used the given data, included units and answered the words in the question?

Practical record

RP 15 / RP 3 · Wire length at constant temperature. Switch off between readings.

Length / cm	p.d. / V	Current / A	$R = V/I$ / ohm
10			
20			
30			
40			
50			
60			
70			
80			
90			
100			

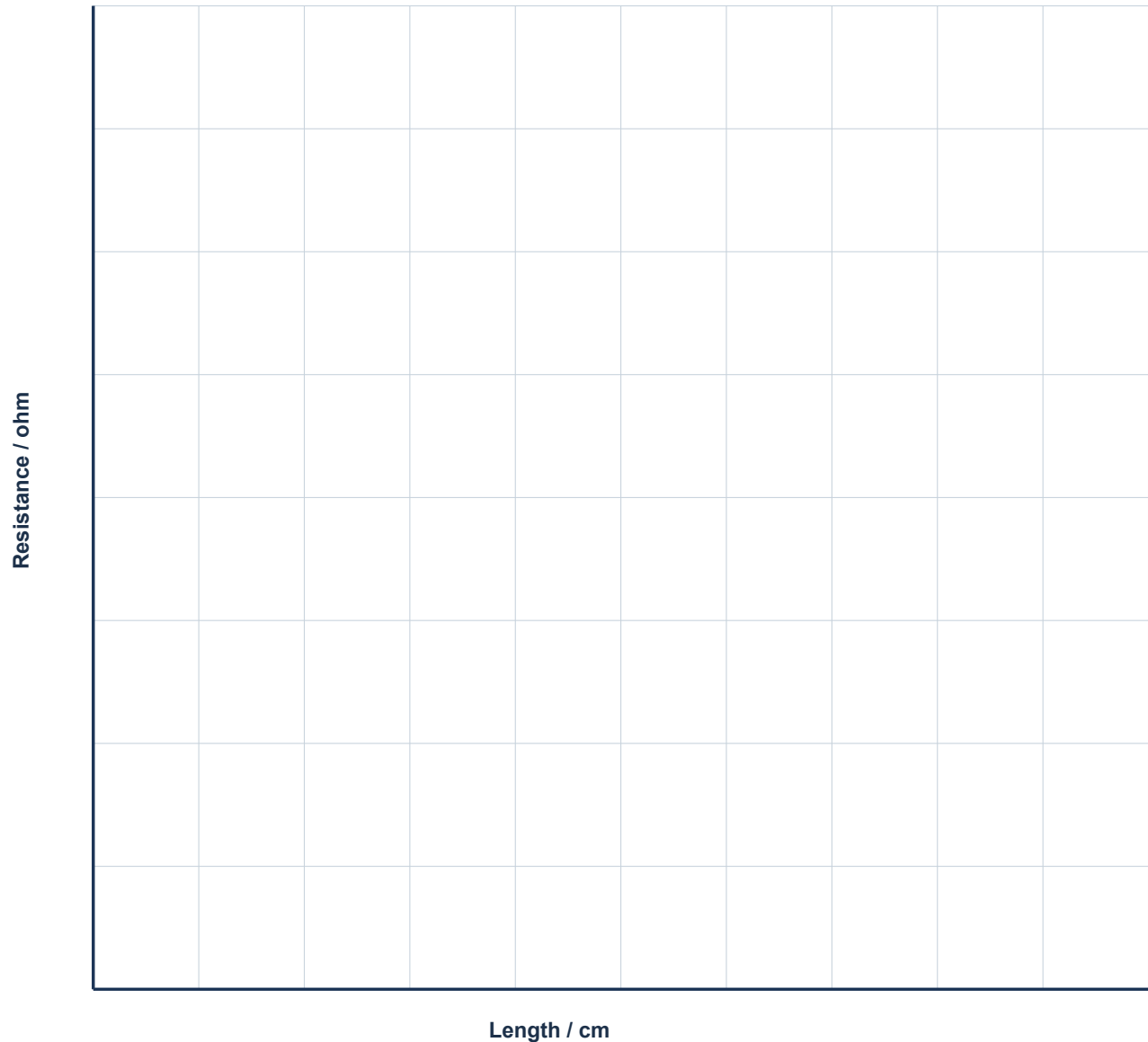
EXPLAIN THE PATTERN

Describe the trend in your wire results. Name one control variable and explain how you kept it constant.

CONCLUSION / EVALUATION

Resistance against length

Use your Lesson 4 data. Choose a sensible scale; plot small crosses; draw a line of best fit.



WHAT DOES YOUR GRAPH SHOW?

Series and parallel

Current, p.d. and resistance rules

THE CASE

One broken lamp turns off a series chain. A parallel branch can keep the other lamp working.

YOUR MISSION

Solve series and parallel circuit questions with clear reasons.

KEY IDEA

Series: current equal, p.d.s add, resistances add. Parallel: p.d. equal across branches, branch currents add; total resistance falls when a branch is added.

WORKED EXAMPLE

2.0 ohm and 4.0 ohm in series on 12 V: $R_{\text{total}} = 6.0 \text{ ohm}$; $I = 2.0 \text{ A}$; p.d.s = 4.0 V and 8.0 V.

1 Build the idea

1. Draw two lamps in series and two lamps in parallel. Mark junctions clearly.

2. Calculate the total resistance and current for 3.0 ohm and 9.0 ohm in series across 12 V.

Measure, reason, answer

2 Use the evidence

3. Parallel branches carry 0.20 A and 0.35 A on a 6.0 V battery. State each branch p.d. and the supply current.

4. Explain, without a parallel-resistance formula, why adding a branch lowers total resistance.

3 AQA-style checkpoint

[5 marks] A 4.0 ohm and an 8.0 ohm resistor are in series on 12 V. Find total resistance, current and p.d. across each. State one parallel-circuit rule.

CHECK BEFORE MOVING ON

Have I shown a formula or rule, used the given data, included units and answered the words in the question?

Resistance in combinations

Required practical: series and parallel, Part B

THE CASE

Adding a second resistor changes the current drawn from the same supply. The direction of change depends on the wiring.

YOUR MISSION

Measure equivalent resistance in two arrangements and explain the results.

KEY IDEA

Measure source V and total I for one resistor, two in series and two in parallel. $R_{\text{total}} = V/I$ each time; series is larger and parallel is smaller than one resistor.

WORKED EXAMPLE

At 6.0 V: one 10 ohm resistor draws 0.60 A; two series draw 0.30 A, so $R_{\text{total}} = 20$ ohm; two parallel draw 1.20 A, so $R_{\text{total}} = 5.0$ ohm.

1 Build the idea

1. Draw all three arrangements and predict the ranking of their total currents.

2. Build each low-voltage circuit. Measure source p.d. and total current; calculate R_{total} in the table.

Measure, reason, answer

2 Use the evidence

3. Explain why the two-series value is higher and the two-parallel value is lower than a single resistor.

4. Evaluate: identify a source of measurement uncertainty and a change that would improve the comparison.

3 AQA-style checkpoint

[5 marks] Two identical resistors each have resistance 12 ohm. Find their series resistance. A student measures 6.0 ohm for their parallel combination. Explain why this is plausible and state one control for a fair comparison.

CHECK BEFORE MOVING ON

Have I shown a formula or rule, used the given data, included units and answered the words in the question?

Practical record

RP 15 / RP 3 · Use the same two resistors and the same supply for each arrangement.

Arrangement	p.d. / V	Total I / A	Rtotal / ohm
One resistor			
Two in series			
Two in parallel			

EXPLAIN THE PATTERN

Rank the three total resistances. Use current, p.d. and the number of paths to justify your ranking.

CONCLUSION / EVALUATION

Reading I-V graphs

Ohmic conductors, lamps and diodes

THE CASE

Three components share the same axes but produce very different curves. Their shape tells you how their resistance behaves.

YOUR MISSION

Recognise characteristic graph shapes and explain what they show.

KEY IDEA

On I (vertical) against V (horizontal): fixed resistor is a straight line through origin at constant temperature; lamp curve flattens as it heats; diode conducts mainly one way.

WORKED EXAMPLE

A graph point is $V = 4.0 \text{ V}$, $I = 0.20 \text{ A}$. At that point $R = 4.0/0.20 = 20 \text{ ohm}$. Do not infer R is constant from one point.

1 Build the idea

1. On a blank set of axes, sketch and label the three characteristic shapes in both directions.

2. Explain why a filament lamp graph becomes less steep as p.d. rises.

Measure, reason, answer

2 Use the evidence

3. At 2.0 V a component carries 0.10 A; at 4.0 V it carries 0.15 A. Calculate R at each point and decide whether it is ohmic.

4. Describe what reversing a diode does to its current.

3 AQA-style checkpoint

[4 marks] A lamp carries 0.20 A at 2.0 V and 0.30 A at 6.0 V. Calculate resistance at each point and explain the change.

CHECK BEFORE MOVING ON

Have I shown a formula or rule, used the given data, included units and answered the words in the question?

Investigating I-V characteristics

Required practical: resistor, lamp and diode

THE CASE

Changing only the test component while keeping the measuring circuit clear lets you compare its electrical behaviour.

YOUR MISSION

Collect bipolar I-V data safely, graph it and justify the three shapes.

KEY IDEA

Vary p.d., record current, reverse the supply for negative readings. Keep a fixed resistor cool; protect the diode with a series resistor; do not exceed the lamp rating.

WORKED EXAMPLE

For a fixed resistor, $V = +3.0 \text{ V}$ and $I = +0.15 \text{ A}$ gives $R = 20 \text{ ohm}$. Reversed readings may be -3.0 V and -0.15 A .

1 Build the idea

1. Draw a series supply-switch-ammeter-variable resistor-test component loop and voltmeter across the test component.

2. Collect readings for fixed resistor, lamp and protected diode, including reverse direction. Record negative signs correctly.

Measure, reason, answer

2 Use the evidence

3. Plot I (y) against V (x) for each device and label the origin and units.

4. Explain the shapes with constant temperature, filament heating and one-way diode conduction.

3 AQA-style checkpoint

[6 marks] Describe a method to obtain the complete I-V characteristic of a filament lamp and state two ways the method changes for a diode.

CHECK BEFORE MOVING ON

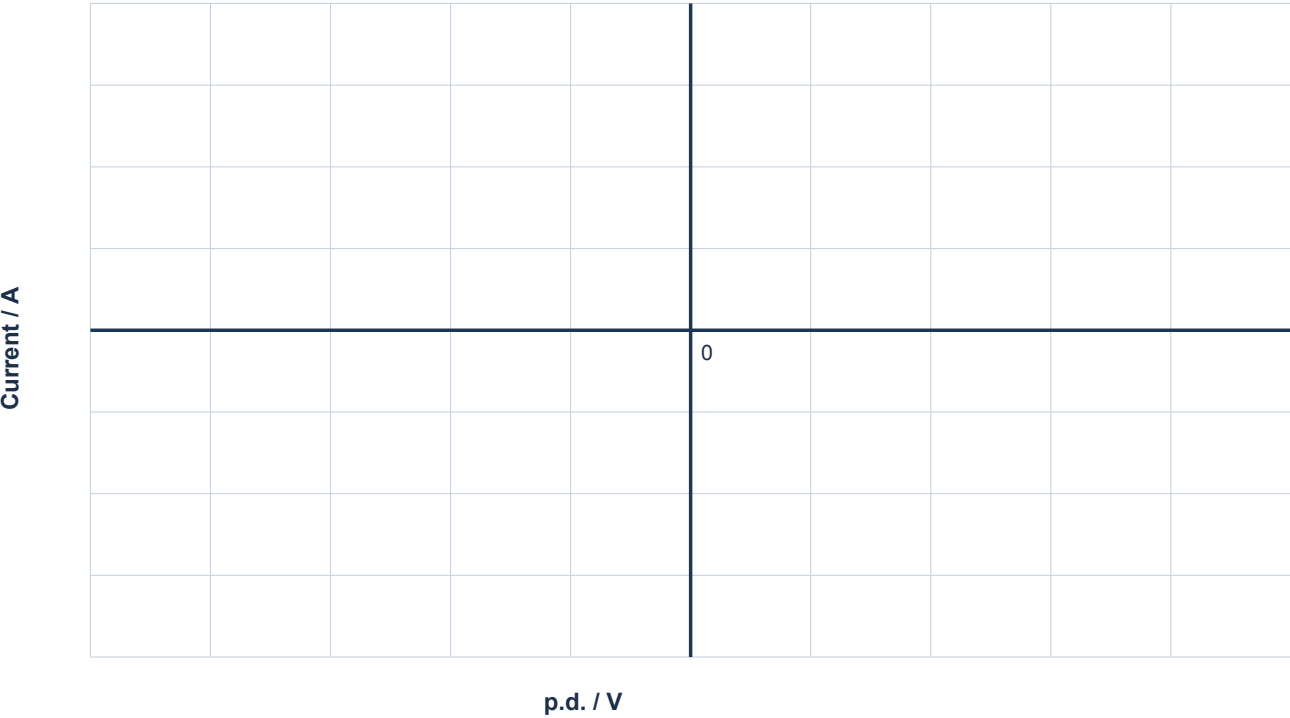
Have I shown a formula or rule, used the given data, included units and answered the words in the question?

Practical record

RP 16 / RP 4 · Use the full practical sheet for every reading and all three graphs. Summarise one point in each direction here; protect the diode.

Device / direction	p.d. / V	Current / A	R at point / ohm
Resistor forward			
Resistor reversed			
Lamp forward			
Lamp reversed			
Diode forward			
Diode reversed			

GRAPH THE FIXED RESISTOR HERE; USE THE FULL SHEET FOR LAMP AND DIODE



CONCLUSION / EVALUATION

Sensors and circuit choices

Thermistors, LDRs and I-V interpretation

THE CASE

An automatic night light and a temperature alarm use resistors whose resistance responds to the surroundings.

YOUR MISSION

Predict how light and temperature change current in simple sensor circuits.

KEY IDEA

NTC thermistor: temperature up, resistance down. LDR: light intensity up, resistance down. At fixed supply p.d., lower total resistance means higher current.

WORKED EXAMPLE

An LDR circuit has 6.0 V and total $R = 600 \text{ ohm}$ in bright light: $I = 6.0/600 = 0.010 \text{ A}$. In dark $R = 1200 \text{ ohm}$: $I = 0.0050 \text{ A}$.

1 Build the idea

1. Sketch R against temperature for an NTC thermistor and R against light intensity for an LDR.

2. Explain the current change in a series circuit containing a thermistor when it gets warmer.

Measure, reason, answer

2 Use the evidence

3. Calculate I for a 12 V sensor circuit with total $R = 300\ \Omega$, then $600\ \Omega$.

4. Choose thermistor or LDR for a fire warning and a streetlamp; justify each choice.

3 AQA-style checkpoint

[4 marks] A 6.0 V supply is connected to an LDR and a fixed resistor in series. In dim light the total resistance is $1500\ \Omega$. Calculate current. Explain what happens to current when the light becomes brighter.

CHECK BEFORE MOVING ON

Have I shown a formula or rule, used the given data, included units and answered the words in the question?

Mains and electrical safety

ac/dc, cable colours and earthing

THE CASE

A metal-cased appliance develops a live-to-case fault. The earth path allows a large fault current so the protection disconnects the supply.

YOUR MISSION

Explain mains hazards and how earthing and fuses protect users.

KEY IDEA

UK mains: ac, about 230 V, 50 Hz. Live brown; neutral blue; earth green/yellow. Live remains dangerous even if a switch is open.

WORKED EXAMPLE

A 50 Hz supply completes 50 cycles each second. A dc supply has a fixed direction; an ac supply repeatedly reverses direction.

1 Build the idea

1. Draw and label a dc p.d.-time line and an ac p.d.-time wave, including axes.

2. Colour-code and state the purpose of live, neutral and earth wires.

Measure, reason, answer

2 Use the evidence

3. Explain why touching live can shock someone even if the lamp switch is open.

4. Explain the sequence from live-to-metal-case fault to the fuse melting or breaker tripping.

3 AQA-style checkpoint

[5 marks] A kettle has a metal case connected to earth. Explain why its live wire remains dangerous when the switch is open, and how the earth wire protects a user after a fault.

CHECK BEFORE MOVING ON

Have I shown a formula or rule, used the given data, included units and answered the words in the question?

Power and energy

$$P = VI, P = I^2R, E = Pt, E = QV$$

THE CASE

A kettle and a lamp use the same mains supply but transfer energy at different rates. Their power ratings tell you how fast.

YOUR MISSION

Choose the right equation, convert units and solve multi-step appliance problems.

KEY IDEA

Power is energy transferred per second. $P = VI$ and $P = I^2R$; $E = Pt$ and $E = QV$. Convert kW to W and minutes/hours to seconds for joules.

WORKED EXAMPLE

2.0 kW kettle for 3.0 min: $P = 2000 \text{ W}$, $t = 180 \text{ s}$, $E = Pt = 360000 \text{ J}$.

1 Build the idea

1. Choose an equation for each pair of known quantities: V and I; I and R; P and t; Q and V.

2. A 230 V heater draws 4.0 A. Calculate its power and energy in 5.0 min.

Measure, reason, answer

2 Use the evidence

3. A 3.0 A current passes a 20 ohm resistor. Calculate dissipated power; explain why the current is squared.

4. A device transfers 240 J while 40 C flows. Calculate its p.d.

3 AQA-style checkpoint

[6 marks] A 2.2 kW kettle runs at 230 V for 150 s. Calculate current and energy transferred in joules. A second kettle draws twice the current through the same resistance: compare its heating power.

CHECK BEFORE MOVING ON

Have I shown a formula or rule, used the given data, included units and answered the words in the question?

The National Grid and exam clinic

Grid efficiency, HT transformer equation and synoptic exam

THE CASE

A power station sends energy over long cables. A step-up transformer raises p.d. so the same power can be transmitted with a smaller current.

YOUR MISSION

Explain efficient transmission and tackle mixed AQA-style electricity questions.

KEY IDEA

For the same transmitted power, raising V lowers I ($P = VI$); smaller I greatly reduces cable heating ($P = I^2R$). Step down before domestic use.

WORKED EXAMPLE

HT: ideal transformer has 25 kV, 400 A in and 400 kV out. $I_s = V_p I_p / V_s = 25$ A.

1 Build the idea

1. Draw a flow from power station to step-up transformer, transmission cables, step-down transformer and home; label why each step matters.

2. Use $P = VI$ to compare currents for 1.0 MW transmitted at 10 kV and 100 kV; then compare I^2R loss for the same cable R .

Measure, reason, answer

2 Use the evidence

3. HT: calculate secondary current for an ideal transformer with 20 kV, 500 A input and 200 kV output.

4. Start the end-of-unit exam clinic in class. Complete remaining questions at home, then correct in a different colour using the mark guidance.

3 AQA-style checkpoint

[4 marks] Explain why the National Grid transmits electricity at high p.d. and why the p.d. is reduced before homes.

CHECK BEFORE MOVING ON

Have I shown a formula or rule, used the given data, included units and answered the words in the question?

End-of-unit exam clinic

Write the equation or rule first. Show each substitution and unit. For explanations, link cause to effect.

1 · CHARGE

A 0.80 A current flows for 2.5 min. Calculate charge. [3]

2 · VOLTAGE

A component transfers 48 J while 8.0 C passes. Calculate its p.d. and explain its meaning. [4]

3 · CIRCUITS

Two 6.0 ohm resistors are in series on 12 V. Find the total resistance, current and p.d. across one resistor. [4]

4 · REQUIRED PRACTICAL

A wire-resistance graph has one point above the trend. Give two plausible causes and describe how you would check it. [4]

End-of-unit exam clinic

Write the equation or rule first. Show each substitution and unit. For explanations, link cause to effect.

5 · I-V

A lamp carries 0.20 A at 2.0 V and 0.30 A at 6.0 V. Calculate R at both points and explain the change. [4]

6 · SAFETY

Explain why a metal-cased appliance needs an earth connection, including the role of a fuse or breaker. [5]

7 · POWER

A 1.5 kW heater runs for 4.0 min on a 230 V supply. Calculate its energy transfer and current. [5]

8 · GRID

Explain high-p.d. transmission using two electrical equations. HT: an ideal transformer has 20 kV, 500 A input and 200 kV output; find output current. [7]

Static electricity

Separate Physics 4.2.5 only · Complete as an extension or independent study after Lesson 12.

KEY IDEAS

Rubbing two insulators can transfer electrons. Gaining electrons makes an object negative; losing electrons leaves it positive. Like charges repel; unlike charges attract. These are non-contact forces.

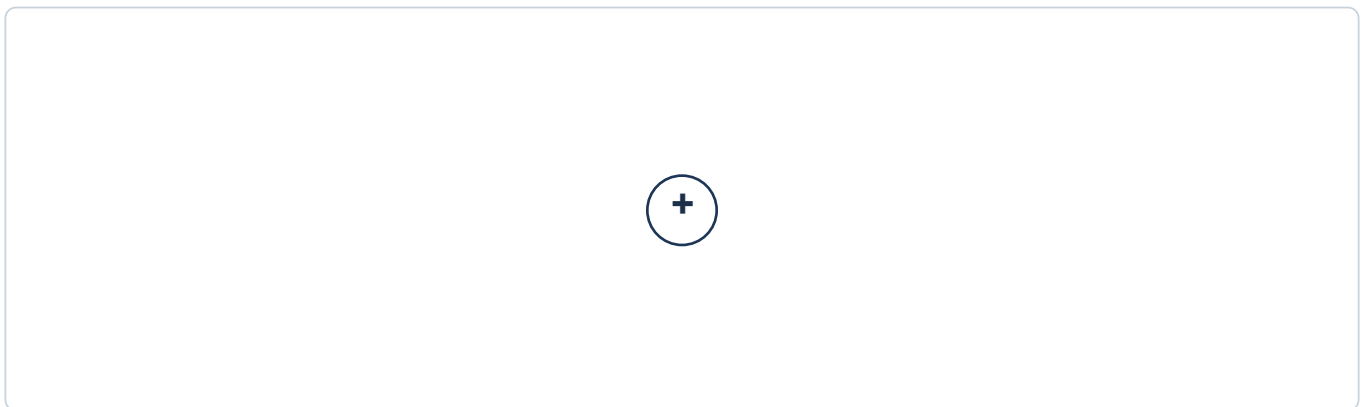
A charged object has an electric field. The field is strongest near it and weaker further away; another charge in the field experiences a force. Sparking can occur when charge moves through the air.

1 · ELECTRON TRANSFER

A plastic rod becomes negative when rubbed with a cloth. Which way did electrons move? What charge is left on the cloth?

2 · DRAW THE FIELD

In the box, draw the field around an isolated positively charged sphere. Add arrows and show where the field is strongest.



3 · AQA-STYLE EXPLANATION [4 MARKS]

A small negative sphere is moved towards a charged positive sphere. Explain the changing force without describing any contact.

Your progress

At the end of each lesson, tick one box honestly. Revisit the pages where you are still developing confidence.

01 Explain what moves in a closed circuit and calculate charge flow.

☐ ☐ ☐

02 Explain p.d. as energy transferred per coulomb and calculate $E = QV$.

☐ ☐ ☐

03 Calculate resistance and explain why a fixed resistor can obey $V = IR$.

☐ ☐ ☐

04 Plan, collect and graph resistance against wire length while controlling temperature.

☐ ☐ ☐

05 Solve series and parallel circuit questions with clear reasons.

☐ ☐ ☐

06 Measure equivalent resistance in two arrangements and explain the results.

☐ ☐ ☐

07 Recognise characteristic graph shapes and explain what they show.

☐ ☐ ☐

08 Collect bipolar I-V data safely, graph it and justify the three shapes.

☐ ☐ ☐

09 Predict how light and temperature change current in simple sensor circuits.

☐ ☐ ☐

10 Explain mains hazards and how earthing and fuses protect users.

☐ ☐ ☐

11 Choose the right equation, convert units and solve multi-step appliance problems.

☐ ☐ ☐

12 Explain efficient transmission and tackle mixed AQA-style electricity questions.

☐ ☐ ☐

NOT YET / GETTING THERE / SOLID

The whole unit in one sentence

Charge flows, energy is transferred, and measurements plus circuit rules explain what happens.